

National University of Computer and Emerging Sciences, Karachi
FAST School of Computing
Re-Midterm II Examination, Fall 2021
14th Dec 2021, 10:00 am to 11:00 am

Course Code: CS3001	Course Name: Computer Networks
Instructor Names: Dr Aqsa Aslam and Mr Shoaib Raza	
Student Roll No:	Section No:

Instructions:-

- Return the question paper.
- **All questions must be answered in the answer script and according to the sequence given in the question paper. There are 5 questions on 2 pages. Each question carries 10 points.**
- In case of any ambiguity, you may make an assumption, however, your assumption should not contradict any statement in the question paper.

Time: 60 minutes

Max Points: 50 points

Question 1: Design a reliable data transfer protocol between the sender and receiver, operating over a channel that can lose or corrupt packets (data or ACKs). Specify the protocol using a finite state machine (FSM) for the sender side of the protocol.

Question 2: Consider a DHCP client-server scenario in figure 1 and answer the following:

- What four pieces of information does a DHCP client obtain from the DHCP server after successful communication between the two?
- What are the destination IP and MAC addresses in the discover request sent by the DHCP client? What is the source IP address and MAC address of the offer message sent by the DHCP server?

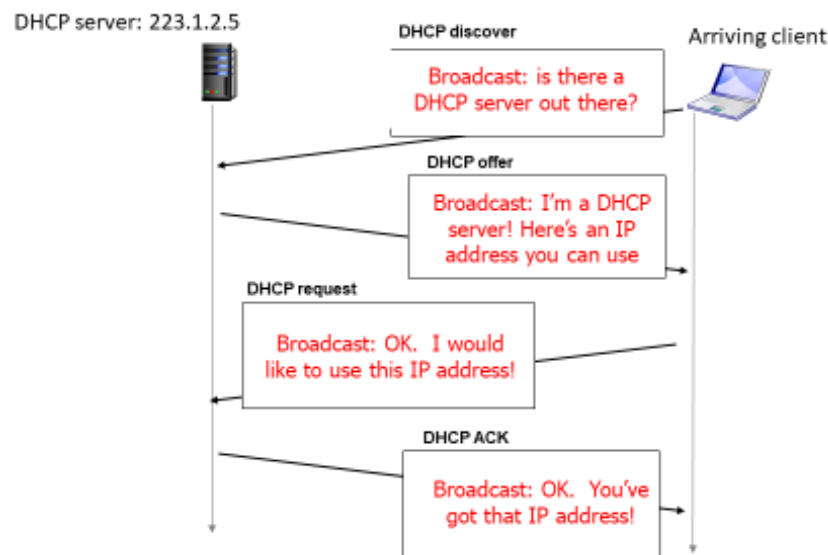


Figure 1 DHCP client-server scenario

Question 3: Draw a timeline diagram for the Selective Repeat algorithm with the sender's window sizes equal to 4. Assume at $t=0$, the sender sends packets 0, 1, 2, 3 and packet 2 will be lost.

Question 4: Your Company wants to utilize the private Class C IP Address of 192.168.1.0. You are tasked with sub-netting the Address to get the most networks with at least 30 Hosts per Subnet. List the following:

- How many Networks will be created after you subnet?
- What is the first usable IP Address in the Second Network range?
- What is the full range of the first three networks?
- What is the subnet mask?

Question 5: Consider figure 2, which plots the evolution of TCP's congestion window at the beginning of each time unit (where the unit of time is equal to the RTT); In the abstract model for this problem, TCP sends a "flight" of packets of size $cwnd$ at the beginning of each time unit. The result of sending that flight of packets is that either (i) all packets are ACKed at the end of the time unit, (ii) there is a timeout for the first packet, or (iii) there is a triple duplicate ACK for the first packet. In this problem, you are asked to reconstruct the sequence of events (ACKs, losses) that resulted in the evolution of TCP's $cwnd$ shown below.

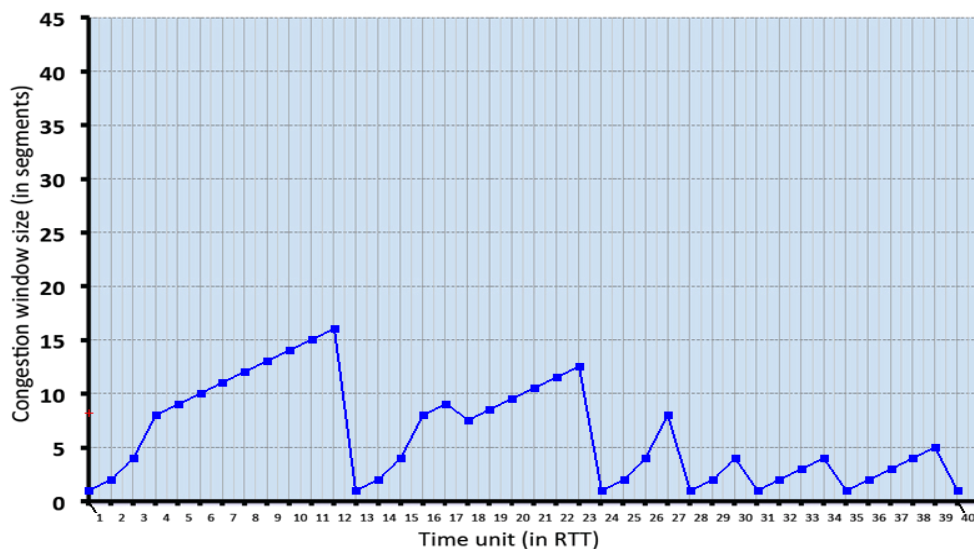


Figure 2 TCP Congestion Control

- Give the time unit(s) (in RTT) at which TCP is in slow start.
- Give the time unit(s) (in RTT) at which TCP is in congestion avoidance.
- Give the time unit(s) (in RTT) at which TCP is in fast recovery.
- Give the time unit(s) (in RTT) at which packets are lost via timeout.
- Give the time unit(s) (in RTT) at which packets are lost via triple ACK.
- Give the time unit(s) (in RTT) at which the value of $ssthresh$ changes (hint: if it changes between $t=3$ and $t=4$, use $t=4$ in your answer)

Best of luck!